

Research Notes on Naive Bayes classifier

Naive Bayes classifier

$$P(C_k | X) \propto P(C_k) \prod_{i=1}^n P(x_i | C_k)$$

>> Section 1

where p_{ki} is the probability of class C_k generating the term x_i . This event model is especially popular for classifying short texts. It has the benefit of explicitly modelling the absence of terms. Note that a naive Bayes classifier with a Bernoulli event model is not the same as a multinomial NB classifier with frequency counts truncated to one.

>> Section 2

$$\begin{aligned} p(C_k, x_1, \dots, x_n) &= p(x_1, \dots, x_n, C_k) = p(x_1 \mid x_2, \dots, x_n, C_k) p(x_2, \dots, x_n, C_k) = p(x_1 \mid x_2, \dots, x_n, C_k) p(x_2 \mid x_3, \dots, x_n, C_k) p(x_3, \dots, x_n, C_k) \\ &= \dots = p(x_1 \mid x_2, \dots, x_n, C_k) p(x_2 \mid x_3, \dots, x_n, C_k) \dots p(x_{n-1} \mid x_n, C_k) p(x_n \mid C_k) p(C_k) \end{aligned}$$

>> Section 3

Thus, the probability ratio $p(S | D) / p(\neg S | D)$ can be expressed in terms of a series of likelihood ratios. The actual probability $p(S | D)$ can be easily computed from $\log(p(S | D) / p(\neg S | D))$ based on the observation that $p(S | D) + p(\neg S | D) = 1$. Taking the logarithm of all these ratios, one obtains:

>> Section 4

$$\text{evidence} = P(\text{male}) p(\text{height} \mid \text{male}) p(\text{weight} \mid \text{male}) p(\text{foot size} \mid \text{male}) + P(\text{female}) p(\text{height} \mid \text{female}) p(\text{weight} \mid \text{female}) p(\text{foot size} \mid \text{female})$$

>> Section 5

$$\frac{p(S \mid D) \, p(\neg S \mid D)}{p(S) \, p(\neg S)} \prod_i \frac{p(w_i \mid S) \, p(w_i \mid \neg S)}{\{ \prod_i \{ p(w_i \mid S) \, p(w_i \mid \neg S) \} \}}$$

>> Section 6

$$p(x=v \mid C_k) = \frac{1}{\sqrt{2\pi} \sigma_k} e^{-\frac{(v-\mu_k)^2}{2\sigma_k^2}}$$

>> Section 7

Probabilistic model Abstractly, naive Bayes is a conditional probability model: it assigns probabilities $p(C_k | x_1, \dots, x_n)$ for each of the K possible outcomes or classes C_k given a problem instance to be classified, represented by a vector $\mathbf{x} = (x_1, \dots, x_n)$ encoding some n features (independent variables). The problem with the above formulation is that if the number of features n is large or if a feature can take on a large number of values, then basing such a model on probability tables is infeasible. The model must therefore be reformulated to make it more tractable. Using Bayes' theorem, the conditional probability can be decomposed as:

>> Section 8

The following example assumes equiprobable classes so that $P(\text{male}) = P(\text{female}) = 0.5$. This prior probability distribution might be based on prior knowledge of frequencies in the larger population or in the training set.

>> Section 9

prior for a given class = $\frac{\text{no. of samples in that class}}{\text{total no. of samples}}$

>> Section 10

However, given the sample, the evidence is a constant and thus scales both posteriors equally. It therefore does not affect classification and can be ignored. The probability distribution for the sex of the sample can now be determined:

>> Section 11

Further reading Domingos, Pedro; Pazzani, Michael (1997). "On the optimality of the simple Bayesian classifier under zero-one loss". *Machine Learning*. 29 (2/3): 103–137. doi:10.1023/A:1007413511361. Webb, G. I.; Boughton, J.; Wang, Z. (2005). "Not So Naive Bayes: Aggregating One-Dependence Estimators". *Machine Learning*. 58 (1): 5–24. doi:10.1007/s10994-005-4258-6.

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Mozina, M.; Demsar, J.; Kattan, M.; Zupan, B. (2004). Nomograms for Visualization of Naive Bayesian Classifier (PDF). Proc. PKDD-2004. pp. 337–348. Archived from the original (PDF) on 2023-11-29. Retrieved 2014-04-01. Maron, M. E. (1961). "Automatic Indexing: An Experimental Inquiry". Journal of the ACM. 8 (3): 404–417. doi:10.1145/321075.321084. hdl:2027/uva.x030748531. S2CID 6692916. Minsky, M. (1961). Steps toward Artificial Intelligence. Proc. IRE. Vol. 49. pp. 8–30.

>> Section 12

The left-hand side of this equation is the log-odds, or logit, the quantity predicted by the linear model that underlies logistic regression. Since naive Bayes is also a linear model for the two "discrete" event models, it can be reparametrised as a linear function $b + \mathbf{w}^T \mathbf{x} > 0$. Obtaining the probabilities is then a matter of applying the logistic function to $b + \mathbf{w}^T \mathbf{x}$, or in the multiclass case, the softmax function. Discriminative classifiers have lower asymptotic error than generative ones; however, research by Ng and Jordan has shown that in some practical cases naive Bayes can outperform logistic regression because it reaches its asymptotic error faster.

>> Section 13

$$\Pr'(S|W) = \frac{s \cdot \Pr(S) + n \cdot \Pr(S|W)}{s + n \Pr(S|W)}$$

>> Section 14

$$p(\neg S | D) = \frac{p(\neg S)}{p(D)} \prod_i p(w_i | \neg S)$$